Afya Smart

AI-powered triage & queue management for fairer, faster public healthcare

Any phone. No app. No internet. USSD *384#

RIDC PT Innovation Hackathon 2026 · Software (Web & Mobile)

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The problem: public clinics are slow — and unfairly so

Tanzania · 1 doctor per ~10,000 people; most citizens own a basic phone

- **Long, opaque queues.** Patients arrive at 5am, wait 4–6 hours, and never know their position or who is being seen before them.
- **No triage before arrival.** A stroke and a refill prescription wait in the same line; the emergency case is discovered too late.
- **Smartphone-only digital health.** Apps and QR check-in exclude the majority who use 2G feature phones.
- **Hidden bias.** Women, rural patients and walk-ins systematically wait longer — nobody measures it, so nobody fixes it.

Solution: triage before you leave home, queue before you arrive

- **USSD on any phone.** Patient dials *384#, answers a short symptom interview in Kiswahili/English — no smartphone, no internet, no app install.
- **AI urgency triage.** ML model (gradient boosting) + rule-based red-flag safety net returns RED / YELLOW / GREEN with a reason.
- **Smart booking.** Nearest facility, first free slot with predicted wait — confirmed by SMS with an appointment ID.
- **Fair live queue.** One queue per facility on the nurse's dashboard: urgency-banded FIFO, emergency bypass, ageing override.
- **Accountability.** A fairness report compares average waits by sex, age, facility and urgency — bias becomes visible and fixable.

How it works — the patient journey

1. Dial *384#

USSD session, Kiswahili or English

2. Describe symptoms

Keyword interview → structured features

3. Triage result

RED / YELLOW / GREEN + reason + advice

4. Book a slot

Facility choice, date, time — capacity-aware

5. SMS confirmation

Appointment ID + reminder before the day

6. Fair queue on arrival

Position on nurse dashboard, live updates

Total time on the phone: under 2 minutes. Walk-ins and anonymous patients (no SIM) supported.

AI triage: ML where it helps, rules where it must never fail

- **Safety net first.** Rule-based red-flag detection (chest pain, stroke signs, bleeding…) always overrides the model.
- **Gradient-boosting classifier** trained on 12,000 synthetic symptom profiles (ml-v2).
- **Clinician feedback loop.** Nurse records the real urgency after consultation → measured against prediction → retraining.
- **Graceful degradation.** Model load failure falls back to rules-only; every prediction carries a version + reason.

89.4%

Overall accuracy (ml-v2)

88.4%

Recall on EMERGENCY cases

12,000

Training samples / version

107

Automated tests passing

Fairness by design — the feature hackathons rarely ship

- **Measured, not assumed.** Average and 90th-percentile waits grouped by sex × age band × facility × urgency × walk-in vs booked.
- **Ageing override.** priority = urgency_weight × 60 − minutes_waited — a GREEN patient who waited long overtakes a fresh YELLOW.
- **Emergency bypass is audited.** Every bypass is flagged and reviewable — speed never erases accountability.
- **Clinician overrides kept.** Nurses can re-grade urgency; the override is stored with who/when/why.
- **Open report.** /fairness renders the numbers as charts anyone — including patients — can inspect.

Technology stack

Built exactly as proposed — Python-first, deployable on lightweight hosting

Patient access (no smartphone needed)	USSD & SMS via Africa's Talking API
Backend / API	Python — FastAPI (SQLAlchemy, Alembic migrations)
AI / ML	Python — gradient-boosting triage classifier + time-series slot-demand forecast
Database	PostgreSQL (production) / SQLite (offline demo) — same models
Facility dashboard	React.js, mobile-first for low-end Android tablets/phones
Patient app	Installable PWA (works on 2G, no app-store dependency)
Hosting	Lightweight cloud (uvicorn) behind Cloudflare tunnel — small payloads on weak links

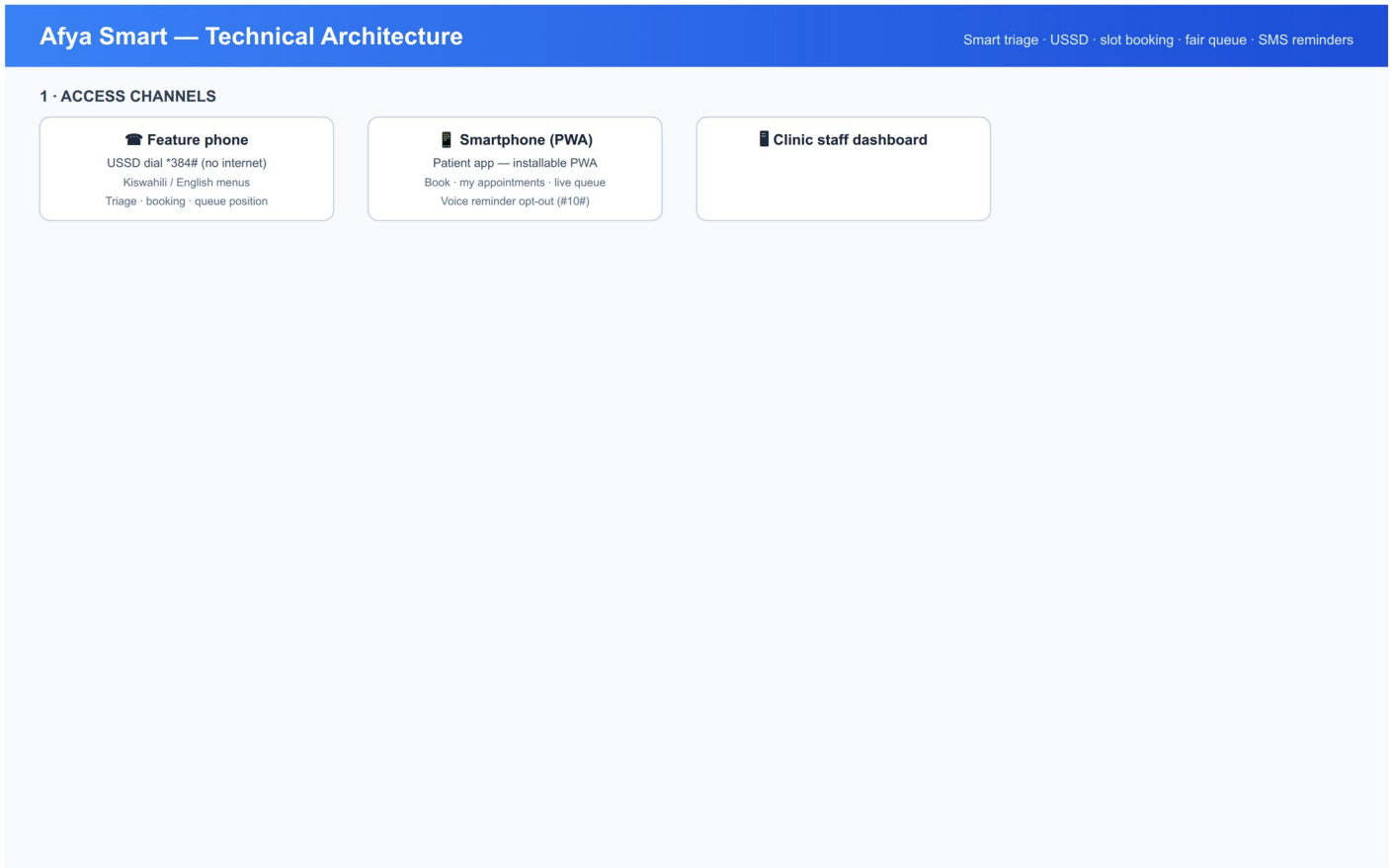
Technical architecture

Single FastAPI service · SQLite/PostgreSQL · Africa's Talking · runs offline-first

This page contains the following errors:

```
error on line 37 at column 89: Unexpected token inside entity:
```

Below is a rendering of the page up to the first error.



Working prototype — end to end, today

6

USSD menus: triage, book, my appointments, facility search, health guide, opt-out

5

Seeded facilities across Dodoma, Morogoro & Singida regions

4

Apps in one: USSD · patient PWA · nurse dashboard · SMS reminders

2

Channels for USSD: live Africa's Talking callback + built-in simulator

- **Full data model:** patients (incl. anonymous), facilities + services, slots, appointments, queue entries, triage sessions — all Alembic-migrated.
- **Integrations live:** Africa's Talking USSD + SMS reminders with dedup, Cloudflare tunnel for a public HTTPS URL on demo day.
- **Deployable anywhere:** uvicorn + SQLite works on a laptop with no internet — important for rural clinics.
- **Honest prototype:** clearly labelled not-for-clinical-use; triage rules need medical sign-off before real patients.

Impact — why this matters for Tanzania

- **Cuts waiting time.** Pre-arrival triage + booked slots mean stable, predictable visits instead of 5am queues.
- **Catches emergencies earlier.** RED flags surface before the patient joins the line — the nurse sees them first on the dashboard.
- **Includes everyone.** Feature phones, walk-ins and SIM-less patients use the same fair system.
- **Gives clinics evidence.** Forecasts demand per slot; the fairness report turns equity from a slogan into a metric.
- **Scales by SMS.** No app downloads, no data bundles — the channel Tanzania already uses for everything.

Success looks like

The exact metrics from our proposal — measured before/after the pilot

- **Reduced average physical wait time.** Baseline vs post-pilot, measured in minutes at the facility (analytics already records every wait).
- **Correct deferral rate.** % of USSD triage sessions that correctly deferred non-urgent cases to self-care or a booked slot — validated against clinician review (feedback loop is live).
- **Flatter demand curve.** Smaller gap between peak-hour and off-peak volume after slot adoption — the forecast shows expected wait per slot to steer patients off the 6–9 AM peak.
- **Adoption rate.** Patients per week using USSD booking vs walk-in, plus repeat usage (tracked per patient).
- **Zero missed emergencies.** No case where a true emergency was delayed by triage or queue flow — audited via emergency-bypass flags and facility incident review.

Roadmap

- **Now (hackathon).** USSD triage + booking, nurse dashboard, fairness report, SMS reminders — all working.
- **Next 3 months.** Pilot with 2–3 public dispensaries; medical board review of triage rules; voice (IVR) menu for low literacy.
- **Next 6 months.** Retrain triage on real clinician feedback; integrate CHAT/WhatsApp channel; county-level demand dashboards.
- **12 months.** District health authority deployments; open fairness metrics published per facility.

The ask

- **Pilot partners.** Public clinics willing to run a 90-day pilot with the nurse dashboard and USSD line.
- **Clinical review.** Medical professionals to sign off (and challenge) the triage rules — a stated condition of this prototype.
- **Mentorship on compliance.** Tanzania Medicines and Medical Devices Authority (TMDA) and data-protection (PDPA) pathways.
- **Follow-on support.** Compute credits and Africa's Talking credits to keep the demo line alive past the hackathon.

Afya Smart — because your place in the queue should never depend on who you are.

github.com/emmanuelbarnabas/afya-smart · demo at /demo on the live server